

ANALOGNETS: ML-HW CO-DESIGN OF NOISE-ROBUST TINYML MODELS AND ALWAYS-ON ANALOG COMPUTE-IN-MEMORY ACCELERATOR

Chuteng Zhou¹ Fernando Garcia Redondo² Julian Büchel^{3,4} Irem Boybat³ Xavier Timoneda Comas³
S. R. Nandakumar³ Shidhartha Das² Abu Sebastian³ Manuel Le Gallo^{*3} Paul N. Whatmough^{*1}

ABSTRACT

Always-on TinyML perception tasks in IoT applications require very high energy efficiency. Analog compute-in-memory (CiM) using non-volatile memory (NVM) promises high efficiency and also provides self-contained on-chip model storage. However, analog CiM introduces new practical considerations, including conductance drift, read/write noise, fixed analog-to-digital (ADC) converter gain, etc. These additional constraints must be addressed to achieve models that can be deployed on analog CiM with acceptable accuracy loss. This work describes *AnalogNets*: TinyML models for the popular always-on applications of keyword spotting (KWS) and visual wake words (VWW). The model architectures are specifically designed for analog CiM, and we detail a comprehensive training methodology, to retain accuracy in the face of analog non-idealities, and low-precision data converters at inference time. We also describe AON-CiM, a programmable, minimal-area phase-change memory (PCM) analog CiM accelerator, with a novel layer-serial approach to remove the cost of complex interconnects associated with a fully-pipelined design. We evaluate the AnalogNets on a calibrated simulator, as well as real hardware, and find that accuracy degradation is limited to 0.8%/1.2% after 24 hours of PCM drift (8-bit) for KWS/VWW. AnalogNets running on the 14nm AON-CiM accelerator demonstrate 8.58/4.37 TOPS/W for KWS/VWW workloads using 8-bit activations, respectively, and increasing to 57.39/25.69 TOPS/W with 4-bit activations.

1 INTRODUCTION

The origins of deep learning were with high-performance GPU hardware consuming hundreds of Watts of power and costing thousands of dollars. However, there is currently great interest in developing networks that can be deployed for inference tasks on constrained devices that can run autonomously on a few milliwatts of power and cost a few dollars, which is often referred to as *TinyML*. The applications for TinyML are numerous and varied, but including things like smart hearing aids (Fedorov et al., 2020), machinery maintenance monitoring (Banbury et al., 2021), human activity recognition (Kodali et al., 2017) and so on. In this paper we focus on two popular always-on perception tasks: visual wake words (VWW) (Chowdhery et al., 2019) and key-word spotting (KWS) (Warden, 2018), which turn on the more sophisticated ML functions when a person is in the camera frame or when a keyword is heard (Figure 1).

In many TinyML scenarios, the limitations of a constrained hardware platform introduce very significant challenges for

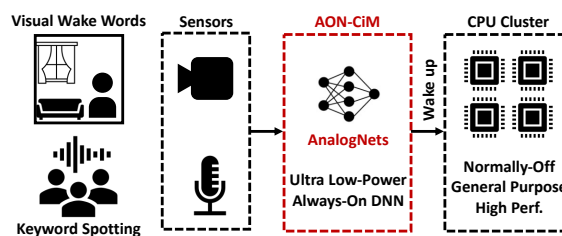


Figure 1. Overview of always-on (AON) TinyML AnalogNets DNNs on self-contained compute-in-memory (CiM) hardware.

DNN deployment. Typical microcontroller (MCU) hardware platforms are cheap ($\sim \$1$) and low power (~ 10 s mW) (Fedorov et al., 2019). However, they often have very limited persistent storage (~ 1 MB), memory (~ 100 s KB) and compute throughput (~ 100 s MOPS) (Banbury et al., 2021). This limits the design of the DNN model in terms of: 1) number of parameters, 2) activation footprint, and 3) number of operations (ops) per inference. These three factors will largely determine the accuracy, latency and energy per inference that can be achieved by a given model.

The high compute and memory demands of DNN inference workloads has recently driven significant research interest in analog compute-in-memory (CiM) hardware platforms (Zidan et al., 2018; Hamdioui et al., 2019; Li et al., 2019; Sebastian et al., 2020). The basic idea is to perform computation

^{*}Equal contribution. ¹Arm Research, Boston ²Arm Research, Cambridge, UK ³IBM Research, Zürich, Switzerland ⁴ETH Zürich, Zürich, Switzerland. Correspondence to: Chuteng Zhou <chu.zhou@arm.com>.